

Winter 2009	CSC343	INTRODUCTION TO DATABASES
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Assignment 3

Due Sunday, March 22, at 11:59pm

The last 48 hours before the deadline are considered a silent period.

Weight: 10% of your final grade

Introduction

For this assignment, you are given a collection of XML files, and its corresponding DTD. The files in this collection represent real data, containing information about clinical trials. For your convenience, we have downloaded this collection from the official government website,

<http://clinicaltrials.gov/>

Please note that the official collection contains over 60,000 files. However, for your assignment you will be using only 10,000 of them. Each of these files contains information about **one** clinical trial.

The collection of the XML files and their corresponding DTD can be downloaded from

<http://www.cs.toronto.edu/~ramona/teaching/csc343/w09/hw3.html>

Data Preparation (20 points)

If you do not intend to solve this assignment on your home computer, you can skip steps 1-2.

1. Download the archive file [hw3_data.zip](#), and the file [trials.dtd](#) from

<http://www.cs.toronto.edu/~ramona/teaching/csc343/w10/hw.html>

2. Extract the 10,000 XML ([NCT00098137.xml](#) to [NCT00997997.xml](#)) files from the archive.
3. **(10 points)** In order to be able to answer the queries in Section 5.6, you should get familiar with the data used for this assignment. This involves understanding the DTD of the XML collection. The information in the following site may help you understand this DTD. We recommend that you read the various information on this site, and try several queries, to get an idea of the content of the trials.

<http://clinicaltrials.gov/ct2/search/advanced>

(Note:) every student who submits some work will get these points. That is, a student who submits nothing, will not get the 10 points.

4. **(10 points)** Write a script (in the scripting language of your choice) to generate a file [collection.xml](#), whose content is as below:

```
<collection>
  <doc href="trials/NCT00000102.xml"/>
  <doc href="trials/NCT00000103.xml"/>
  ...
  <doc href="trials/NCT00064051.xml"/>
</collection>
```

where `NCT00000102.xml`, `...`, `NCT00064051.xml` are XML files from the given collection.

- (a) For each file in the XML collection, there will be a `doc` element, child of the `collection` element. In other words, there will be 10,000 `doc` children of `collection`.
- (b) All the queries that you will write in Section 5.6 will be against the `collection.xml` file.
- (c) To query an XML collection, you will have to use the XQuery `collection` function. For instance, if you want to output everything in the collection, the XQuery query is:

```
collection("collection.xml")
```

The result of this query is a file over 114M in size!

Structure Rules

For all the queries you will write in XQuery (answers to the questions in Section 5.6), you will have to follow these **structure rules**:

1. Each XQuery that you will write has to be stored in a separate file, `qX.xq`, where `X` is a number between 1 and 7.
2. The output of each query must be stored in a separate file, `qX.xml`, where `X` is a number between 1 and 7. For your convenience, you will be provided with the format of each of these files.
3. The statement to output the result of each query in its corresponding output file is the following:

```
saxon-xq -q:queryX.xq -o:qX.xml,
```

where `saxon-xq` is the command (I use) to run Saxon.

If you are using Saxon in Windows, you may use the following file, `saxon-xq.cmd` to run Saxon:

```
@echo off
java -Xmx900M -cp C:\Saxon\saxon9.jar net.sf.saxon.Query %*
```

Content Rules

For all the queries you will write in XQuery (answers to the questions in Section 5.6), you will have to follow these **content rules**:

1. Unless otherwise specified, when a question asks for trials that were “performed”, your query should iterate over **all** the trials in the collection.
2. By “ID of a trial” we refer to the `nct_id` element corresponding to that trial.
3. If you have to eliminate duplicates from your result, you should use the `distinct-values()` function (for more information, please check the documentation).

4. If you need to check that a `PCDATA` element (or part of it) satisfies a certain condition, you can use the `contains()` function. Please see the documentation for more information.
5. In the collection, some elements are empty; consequently, some of the elements in your output will also be empty.
6. For guidance, we have included comments in the “output format” of some queries. You do not have to include these comments in your result.

XQuery Queries (80 points)

Write the following queries in XQuery.

1. (5 points) List all the countries where a clinical trial was performed, in alphabetical order.

The **output** format is:

```
<countries>
  <country>...</country>
  ...
</countries>
```

Note: each country should be listed only once.

2. (5 points) Find all the conditions that are being investigated, in alphabetical order.

The **output** format is:

```
<conditions>
  <condition>...</condition>
  ...
</conditions>
```

Note: each condition should be listed only once.

3. (10 points) For each clinical trial, output its ID, the year when it started and the year when it ended.

The **output** format is:

```
<trials>
  <trial>
    <nct_id>...</nct_id>
    <start_year>...</start_year>
    <end_year>...</end_year>
  </trial>
  ...
</trials>
```

Hint: you will have to do some string manipulations in order to extract the year information (for more information, please check the documentation).

4. (10 points) For each clinical trial that ended in 2008, find the number of distinct countries where it was performed. Your output should also include the ID of each such trial.

The **output** format is:

```

<trials_end_2008>
  <trial>
    <nct_id>...</nct_id>
    <NoOfCountries>...</NoOfCountries>
  </trial>
  ...
</trials_end_2008>

```

Hint: you can use the `contains()` function to check the year information (for more information, please check the documentation).

Write the following queries in XQuery.

5. **(15 points)** For each country where at least one clinical trial was performed, output the number of trials that were performed there.

The **output** format is:

```

<trials_per_country>
  <trial_per_country>
    <country>...</country>
    <NoOfTrials>...</NoOfTrials>
  </trial_per_country>
  ...
</trials_per_country>

```

Hint: you can use `q1.xml` to answer this query.

6. **(15 points)** List the clinical trials performed in more than 1 country. For each such trial, your output should include its ID and all the countries where that trial was performed.

The **output** format is:

```

<trials_more_countries>
  <trial_more_countries> (: one trial and all the countries where it was performed :)
    <nct_id>...</nct_id>
    <countries>
      <country>...</country>
      ... (: all the countries where the trial was performed :)
    </countries>
  </trial_more_countries>
  ...
</trials_more_countries>

```

7. **(20 points)** For each condition being investigated, list all the clinical trials investigating it. For each of the clinical trials associated with a condition, your output should include its ID and the address of all the facilities where that clinical trial is performed.

The **output** format is:

```

<trial_conditions>
  <trial_condition> (: one condition and all the trials associated with it :)
    <condition>...</condition>
    <trials> (: all the trials associated with this condition :)
      <trial>
        <nct_id>...</nct_id>

```

```

    <address>...</address>
    ... (: there can be more facilities where this trial is performed :)
    <address>...</address>
  </trial>
  ... (: more trials :)
</trials>
</trial_condition>
...
</trial_condition>

```

Hint: you can use `x2.xml` to answer this query.

Submission instructions

The submission instructions below apply to both on-time and late submissions.

You can work in groups of 2 people, with only **one** submission per group.

The submission must contain the following files (with these exact names):

1. `group.txt` — text file, with information about **all** members of the group:
 - Last name:
 - First name:
 - Student number:
 - Grace days used:
 - CDF submission account:
2. `q[1-7].xq`, your XQuery queries.
3. There is no need to submit your script.
4. to submit, use the command:


```
submit -c csc343h -a A3 group.txt q1.xq q2.xq q3.xq q4.xq q5.xq q6.xq q7.xq
```
5. If you are using grace days, the name of the folder to submit is `A3graceX`, where X is the number of grace days used (1, 2 or 3).
 - For more information on `submit`, use `man submit`.

Suggestions

XML/XPath/XQuery Useful Information

1. The World Wide Web Consortium (W3C): <http://www.w3.org/>
2. XQuery 1.0: An XML Query Language: <http://www.w3.org/TR/2007/REC-xquery-20070123/>
3. XML Query Implementations: <http://www.w3.org/XML/Query/#implementations>
4. SAXON: <http://saxonica.com/>
 - you can use either `SaxonHE` or `Saxon-B`
5. XML Query Use Cases: <http://www.w3.org/>

Testing your XQuery queries

1. Before testing your queries over the entire collection of files, you may want to test them on one file or very few files only. The largest file may be a potential candidate for testing.
2. Another way to test your queries is using the following website:

<http://clinicaltrials.gov/ct2/search/advanced>

Testing your queries on this site may give an indication of the correction of your queries.

Good luck!